

Absolute Value Inequalities: Special Cases and Interval Notation

TEACHER KEY | Algebra II | Unit 2 | Day 8 | TEK A2.6.F

Objective: I can write absolute value inequality solutions in interval notation, solve harder cases, and recognize when an inequality has no solution.

Vocabulary

No solution — NO value of x works. Happens when an isolated absolute value is set $<$ or $\leq$ a negative number.

Part 1 — Interval Notation Reference

Inequality	Compound form	Interval notation
$ x > 5$	$x < -5$ or $x > 5$	$(-\infty, -5) \cup (5, \infty)$
$ x \geq 5$	$x \leq -5$ or $x \geq 5$	$(-\infty, -5] \cup [5, \infty)$
$ x < 5$	$-5 < x < 5$	$(-5, 5)$
$ x \leq 5$	$-5 \leq x \leq 5$	$[-5, 5]$

Part 2 — Variable on Both Sides

Example. $|3x + 1| - 4 \leq 2x - 1$

Isolate: $|3x + 1| \leq 2x + 3$

$-(2x + 3) \leq 3x + 1 \leq 2x + 3$

Solve both halves and combine: $[-4/5, 2]$

Part 3 — When There Is No Solution

Example. $4 + |x + 1| \leq 2$

Isolate: $|x + 1| \leq -2$

An absolute value can never be negative, so it can never be $\leq$ a negative number. Solution: **no solution**

Practice

- Write $|x| \geq 8$ in interval notation. $(-\infty, -8] \cup [8, \infty)$
- Write $|x| < 3$ in interval notation. $(-3, 3)$
- Solve $|x - 2| + 6 \leq 1$ **no solution**
- Solve $|2x - 1| > x + 5$ $x < -4/3$ or $x > 6$

Exit Ticket

Solve $3 + |x - 4| < 1$, and explain why the answer comes out the way it does.

no solution - isolating gives $|x - 4| < -2$, and an absolute value can never be less than a negative number